MATH121-01 Practice Questions for Exam 2

instructor: John MacCormick

These practice questions and solutions were written by GitHub Copilot, an AI assistant. The AI assistant was provided with homework assignments 5, 6, and 7 and with a copy of the first midterm exam as an example of the correct formatting. The assistant was then given the following prompt:

> Please use the format of the attached MATH121 exam 1 to create a new set of practice questions in the empty file practice-questions-exam2. The topics under consideration are all given in the attached homework folder.
>
> Please choose a representative sample of questions that cover similar topics to those in the homework.

The instructor has attempted to verify the appropriateness of the questions and the correctness of the solutions. The instructor has also made minor edits.
Questions for the real exam are written by the instructor *without* AI assistance. These practice questions were written *with* AI assistance for two reasons: (i) to save time and effort in providing a substantial number of practice questions; (ii) to provide a demonstration of how easy it is to generate additional practice questions using AI tools, which is permissible and encouraged for this class.

Question 1. Probability and conditional probability

Among 120 students surveyed about two streaming services, 52 had watched Service A, 60 had watched Service B, and 24 had watched both services. One student was selected at random.

(a) Find the probability that the student had watched Service A or Service B (or both).

(b) Find the probability that the student had watched neither service.

(c) Given that the student had watched Service B, find the probability that the student had also watched Service A.

(d) Are the events "watched Service A" and "watched Service B" independent? Justify your answer.

(e) Are these events mutually exclusive? Explain.

Solution.

(a) Using $P(A \cup B) = P(A) + P(B) - P(A \cap B)$,

$$P(A \cup B) = \frac{52 + 60 - 24}{120} = \frac{88}{120} \approx 0.733.$$

(b) The probability of neither is $1 - P(A \cup B) = 1 - \frac{88}{120} = \frac{32}{120} \approx 0.267$.

(c) $P(A \mid B) = P(A \cap B)/P(B) = \frac{24/120}{60/120} = 0.4$.

(d) No. We have $P(A \mid B) = 0.4$, whereas $P(A) = \frac{52}{120} \approx 0.433$. Since these are unequal, the events are dependent.

(e) No. The events are not mutually exclusive because 24 students watched both.

Question 2. Discrete probability distribution

Let X be the number of online orders received by a small shop during a randomly selected hour. The probability distribution is shown below.

x	0	1	2	3
$P(X = x)$	0.10	0.25	0.40	0.25

(a) Find the probability that the shop receives at least two orders in an hour.

(b) Find the expected number of orders per hour.

(c) Given that at least one order arrives, find the probability that at least two orders arrive.

Solution.

(a) $P(X \geq 2) = P(X = 2) + P(X = 3) = 0.40 + 0.25 = 0.65$.

(b)
$$E(X) = 0(0.10) + 1(0.25) + 2(0.40) + 3(0.25) = 1.80.$$

(c)
$$P(X \geq 2 \mid X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)} = \frac{0.65}{0.25 + 0.40 + 0.25} = \frac{13}{18} \approx 0.722.$$

Question 3. Normal distribution

The amount of coffee dispensed by a machine into a cup is approximately normally distributed with mean 500 milliliters and standard deviation 12 milliliters.

(a) What proportion of cups contain less than 482 milliliters?

(b) What proportion of cups contain between 494 and 518 milliliters?

(c) The machine is adjusted so that only the cups in the top 10% by volume exceed a particular amount. What is that volume cutoff?

Solution.

Let X be the amount dispensed, so $X \sim N(500, 12)$, where the second value is the standard deviation.

(a)
$$z = \frac{482 - 500}{12} = -1.5, \qquad P(X < 482) = P(Z < -1.5) \approx 0.0668.$$

(b) The corresponding z-scores are -0.5 and 1.5. Thus
$$P(494 < X < 518) = P(-0.5 < Z < 1.5) \approx 0.9332 - 0.3085 = 0.6247.$$

(c) The 90th percentile of the standard normal distribution is approximately $z = 1.28$ (corresponds to 0.8997 on our table). The cutoff is
$$500 + 1.28(12) = 515.36,$$
or approximately 515.4 milliliters.

Question 4. Uniform distribution

The waiting time, X minutes, for a shuttle bus to arrive is uniformly distributed from 0 to 10 minutes.

(a) Sketch the probability density function, labeling both axes.

(b) Find the probability that a randomly selected passenger waits between 2.5 and 7 minutes.

(c) Find the 40th percentile of the waiting time distribution.

(d) What is the median waiting time?

Solution.

(a) The density is constant at height $1/10 = 0.1$ from $x = 0$ to $x = 10$ and zero elsewhere. The graph is a rectangle with horizontal axis x (waiting time in minutes) and vertical axis $f(x)$ (density).

(b) For a uniform distribution, probability is interval length divided by total length:
$$P(2.5 < X < 7) = \frac{7 - 2.5}{10 - 0} = 0.45.$$

(c) The 40th percentile is 40% of the way from 0 to 10:
$$0 + 0.40(10) = 4 \text{ minutes}.$$

(d) The median is the 50th percentile, so it is 5 minutes.

Question 5. Sampling distribution of a sample proportion

Suppose 12% of all packages shipped by a company arrive later than promised. A random sample of 200 packages is selected. Let $\hat{p}$ be the sample proportion that arrive late.

(a) Verify that the sample size is large enough to use the CLT.

(b) Describe the sampling distribution of $\hat{p}$ by giving its mean and standard deviation.

(c) Find the probability that fewer than 9% of the sampled packages arrive late.

(d) Find the probability that between 10% and 15% of the sampled packages arrive late.

Solution.

Here $p = 0.12$ and $n = 200$.

(a) $np = 200(0.12) = 24$ and $n(1-p) = 200(0.88) = 176$, both at least 10. The CLT is appropriate.

(b) The sampling distribution has mean

$$\mu_{\hat{p}} = p = 0.12$$

and standard deviation

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(0.12)(0.88)}{200}} \approx 0.0230.$$

(c)

$$z = \frac{0.09 - 0.12}{0.0230} \approx -1.31, \qquad P(\hat{p} < 0.09) \approx P(Z < -1.31) \approx 0.095.$$

(d)

$$z_{\text{low}} = \frac{0.10 - 0.12}{0.0230} \approx -0.87, \qquad z_{\text{high}} = \frac{0.15 - 0.12}{0.0230} \approx 1.30.$$

Therefore

$$P(0.10 < \hat{p} < 0.15) \approx P(-0.87 < Z < 1.30) \approx 0.9032 - 0.1922 = 0.7110.$$

Question 6. Confidence interval for a population proportion

In a random sample of 120 residents, 72 said they support a proposed park improvement.

(a) Find the sample proportion, which is the point estimate of the population proportion.

(b) Construct a 95% confidence interval for the population proportion.

(c) Verify that the sample size is large enough for the confidence interval procedure to be valid.

(d) Is a population proportion of 0.70 consistent with this confidence interval? Explain.

(e) If the same sample were used to construct both a 90% and a 95% confidence interval, which interval would have the larger margin of error? Explain.

Solution.

(a)
$$\hat{p} = \frac{72}{120} = 0.60.$$

(b) For 95% confidence, $z^* = 1.96$. The standard error and margin of error are

$$SE = \sqrt{\frac{(0.60)(0.40)}{120}} \approx 0.0447, \qquad ME = 1.96(0.0447) \approx 0.0877.$$

The confidence interval is

$$0.60 \pm 0.0877 = (0.5123,\ 0.6877),$$

or approximately (0.512, 0.688). Note: $z^* = 2$ would also be acceptable for 95% confidence (based on Empirical Rule), giving a slightly wider interval (0.511, 0.689).

(c) There are $\hat{p} \times 120 = 72$ successes and $(1 - \hat{p}) \times 120 = 48$ failures in the sample, both at least 10. The normal-based interval is appropriate.

(d) No. The value 0.70 is outside the interval, so it is not among the population proportions that are reasonably consistent with this interval.

(e) The 95% interval has the larger margin of error. A higher confidence level requires a larger critical value, while the standard error remains the same.

Question 7. Checking confidence intervals

A poll reports a sample proportion of 0.60. Four confidence intervals are proposed, all claimed to have been computed from that same sample:

$$(0.57, 0.63), \qquad (0.60, 0.66), \qquad (0.58, 0.62), \qquad (0.47, 0.53).$$

Which intervals could be correct? Explain.

Solution.

A confidence interval for a proportion, computed by the methods used in class, is centered at the sample proportion, 0.60. The intervals $(0.57, 0.63)$ and $(0.58, 0.62)$ are centered at 0.60 and could be correct. The interval $(0.60, 0.66)$ is centered at 0.63, and $(0.47, 0.53)$ is centered at 0.50; neither could be correct for this sample.